

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the function $f(x) = \sqrt{1-x}$ on the interval $[-9, -1]$.
 - a) Approximate the area under the curve using a midpoint estimation with 4 subintervals.

We have that $\Delta x = \frac{-1 - (-9)}{4} = 2$. Therefore,

$$\begin{aligned} \int_{-9}^{-1} \sqrt{1-x} dx &\approx 2(f(\bar{x}_1) + f(\bar{x}_2) + f(\bar{x}_3) + f(\bar{x}_4)) \\ &= 2(f(-8) + f(-6) + f(-4) + f(-2)) \\ &= 2(\sqrt{9} + \sqrt{7} + \sqrt{5} + \sqrt{3}) \\ &\approx 19.2277 \end{aligned}$$

- b) Approximate the area under the curve using a trapezoidal estimation with 4 subintervals.

$$\begin{aligned} \int_{-9}^{-1} \sqrt{1-x} dx &\approx \frac{2}{2}(f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)) \\ &= 1(f(-9) + 2f(-7) + 2f(-5) + 2f(-3) + f(-1)) \\ &= \sqrt{10} + 2\sqrt{8} + 2\sqrt{6} + 2\sqrt{4} + \sqrt{2} \\ &\approx 19.1323 \end{aligned}$$

- c) Approximate the area under the curve using Simpson's rule with 4 subintervals.

$$\begin{aligned} \int_{-9}^{-1} \sqrt{1-x} dx &\approx \frac{2}{3}(f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + f(x_4)) \\ &= \frac{2}{3}(f(-9) + 4f(-7) + 2f(-5) + 4f(-3) + f(-1)) \\ &= \frac{2}{3}(\sqrt{10} + 4\sqrt{8} + 2\sqrt{6} + 4\sqrt{4} + \sqrt{2}) \\ &\approx 19.1928 \end{aligned}$$

- d) The approximation made in part (a) has some error associated with it which can be estimated by the inequality $|E_M| \leq \frac{K(b-a)^3}{24n^2}$ where $|f''(x)| \leq K$ for all x in $[a, b]$.
- i) Determine a value of K which can be used to help estimate the upper bound on the error.

We note that

$$f'(x) = \frac{-1}{2\sqrt{1-x}}$$

$$f''(x) = \frac{-1}{4(1-x)^{3/2}}$$

In order to find a value of K for which $|f''(x)| \leq K$ we attempt to find the absolute minimum and maximum values of $f''(x)$ on $[-9, -1]$. We use the closed interval method from Calculus I.

Note that $f'''(x) = \frac{-3}{8(1-x)^{5/2}}$.

$f'''(x)$ never equals 0.

$f'''(x)$ is undefined when $x = 1$ (which is outside the interval).

Since no critical points were found we only calculate the outputs of $f''(x)$ at the endpoints of the interval.

$$f''(-9) = \frac{-1}{4(1-(-9))^{3/2}} = \frac{-1}{4(10)^{3/2}} \approx -0.00779$$

$$f''(-1) = \frac{-1}{4(1-(-1))^{3/2}} = \frac{-1}{4(2)^{3/2}} \approx -0.0884$$

So we see that $|f''(x)| \leq 0.09$ and so we let $K = 0.09$

(I chose $K = 0.09$ instead of 0.0884 since we have to divide by 3 in the next part and 0.0884 isn't divisible by 3.

- ii) Determine the upper bound on the error of the approximation from part (a).

We have that $|E_M| \leq \frac{K(b-a)^3}{24n^2} = \frac{0.09(8)^3}{24(4)^2} = \frac{0.09(4)}{3} = 0.12$.

Therefore, we know that the answer from part (a) is, at most, 0.12 units away from the correct answer.

- e) Determine the values of n needed for a midpoint approximation and trapezoidal approximation to estimate the total area under $f(x)$ on $[-9, -1]$ to within $\frac{1}{100}$.

We know the formulas for finding the errors associated with these approximation techniques are

$$|E_T| \leq \frac{K(b-a)^3}{12n^2} \quad \text{and} \quad |E_M| \leq \frac{K(b-a)^3}{24n^2} \quad \text{where } |f''(x)| \leq K \text{ for all } x \text{ in } [a, b].$$

We note that we want $\frac{K(b-a)^3}{12n^2} < \frac{1}{100}$ and $\frac{K(b-a)^3}{24n^2} < \frac{1}{100}$ and we can use $K = 0.09$ which was established in part (d).

Therefore,

$$\frac{K(b-a)^3}{12n^2} < \frac{1}{100} \rightarrow \frac{0.09(8)^3}{12n^2} < \frac{1}{100} \rightarrow n^2 > 384 \rightarrow n > \sqrt{384} \approx 19.6 \rightarrow n \geq 20$$

$$\frac{K(b-a)^3}{24n^2} < \frac{1}{100} \rightarrow \frac{0.09(8)^3}{24n^2} < \frac{1}{100} \rightarrow n^2 > 192 \rightarrow n > \sqrt{192} \approx 13.9 \rightarrow n \geq 14$$

So, we must use at least 28 trapezoids to estimate $\int_{-9}^{-1} f(x) dx$ to within $\frac{1}{100}$ of its actual value.

We must use at least 20 midpoint rectangles to estimate $\int_{-9}^{-1} f(x) dx$ to within $\frac{1}{100}$ of its actual value.

2. Determine the value of n needed for Simpson's rule to estimate the area under a curve $f(x)$ on $[-1, 1]$ to within $\frac{1}{100}$ if we know that $f^{(4)}(x) = 3x^{1/3} - 6x^{4/3}$.

We know the formulas for finding the errors associated with these approximation techniques are

$$|E_S| \leq \frac{K(b-a)^5}{180n^4} \quad \text{where } |f^{(4)}(x)| \leq K \text{ for all } x \text{ in } [a, b].$$

We first find a value of K for which $|f^{(4)}(x)| \leq K$ for all x in $[-1, 1]$.

We calculate $f^{(5)}(x) = x^{-2/3} - 8x^{1/3} = x^{-2/3}[1 - 8x] = \frac{1-8x}{x^{2/3}}$.

$$f^{(5)}(x) = 0 \text{ when } 1 - 8x = 0 \rightarrow x = \frac{1}{8}.$$

$f^{(5)}(x)$ is undefined when $x = 0$.

We test these critical points and the endpoints of the interval.

x	$f^{(4)}(x)$
-1	-9
0	0
1/8	9/8
1	-3

So, we see that $|f^{(4)}(x)| \leq 9$ and so we let $K = 9$.

We note that we want $\frac{K(b-a)^5}{180n^4} < \frac{1}{100}$ and we can use $K = 9$.

Therefore,

$$\frac{K(b-a)^5}{180n^4} < \frac{1}{100} \rightarrow \frac{9(2)^3}{180n^4} < \frac{1}{100} \rightarrow n^4 > 40 \rightarrow n > \sqrt[4]{40} \approx 2.5 \rightarrow n \geq 3$$

So, we must use at least 3 subintervals to estimate $\int_{-1}^1 f(x) dx$ to within $\frac{1}{100}$ of its actual value when using Simpson's method.

3. Does the improper integral $\int_1^{\infty} \sin(x) dx$ converge or diverge? Why?

$$\int_1^{\infty} \sin(x) dx = \lim_{t \rightarrow \infty} \left[\int_1^t \sin(x) dx \right] = \lim_{t \rightarrow \infty} \left[\cos(x) \Big|_1^t \right] = \lim_{t \rightarrow \infty} [\cos(t) - \cos(1)].$$

This limit DNE because the $\lim_{t \rightarrow \infty} \cos(t)$ DNE.

4. Determine whether the integral is convergent or divergent. If convergent, evaluate the integral.

a) $\int_0^1 \frac{1}{\sqrt{x}} dx$

$$\int_0^1 \frac{1}{\sqrt{x}} dx = \lim_{t \rightarrow 0^+} \left[\int_t^1 \frac{1}{\sqrt{x}} dx \right] = \lim_{t \rightarrow 0^+} \left[2\sqrt{x} \Big|_t^1 \right] = \lim_{t \rightarrow 0^+} \left[2 - 2\sqrt{t} \right] = 2 - 0 = 2.$$

Therefore, the integral converges to 2.

b) $\int_0^{\infty} \frac{x^2}{27+x^3} dx$

$$\int_0^{\infty} \frac{x^2}{27+x^3} dx$$

$$\int_0^{\infty} \frac{x^2}{27+x^3} dx = \lim_{t \rightarrow \infty} \left[\int_0^t \frac{x^2}{27+x^3} dx \right]. \text{ Let } u = 27+x^3, \text{ then } du = 3x^2 dx \rightarrow \frac{1}{3} du = x^2 dx. \text{ Our bounds}$$

become $u(0) = 27$ & $u(t) = 27+t^3$. Therefore, we have

$$\begin{aligned} \int_0^{\infty} \frac{x^2}{27+x^3} dx &= \lim_{t \rightarrow \infty} \left[\int_0^t \frac{x^2}{27+x^3} dx \right] = \lim_{t \rightarrow \infty} \left[\int_{27}^{27+t^3} \frac{1}{3} \cdot \frac{1}{u} du \right] = \lim_{t \rightarrow \infty} \left[\frac{1}{3} \ln |u| \Big|_{27}^{27+t^3} \right] \\ &= \lim_{t \rightarrow \infty} \left[\frac{1}{3} \ln |27+t^3| - \frac{1}{3} \ln(27) \right] = \infty - \frac{1}{3} \ln(27) = \infty \end{aligned}$$

Therefore, the integral diverges.